

ZS-F36

The Integral UV Normalization of the Z-Spin Odd Three-Form: Exact M_{eff} Reduction, Wrapped-Brane Charge Pairing on the Koenigs Torus, and the Metric-Scale Gate

Author: Kenny Kang

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Theme / Code: Foundations / ZS-F36 · Companion to ZS-F33 / ZS-F34 / ZS-F35 · Register normalization from ZS-M6 · Cosmological hand-off ZS-A31

Verification: 56/56 (zs_f36_verify_v2_1.py) + 14/14 (zs_f36_review_verify.py) PASS | **Zero fitted parameters** | primitive charge = 1 and dimensionless WZ phase 2π **PROVEN** (Smith form); dimensionful $c_e = 2\pi$ **DERIVED-CONDITIONAL** on $\alpha_{UV} = 1$; $\tilde{G}_s = 1$ **DERIVED-CONDITIONAL** on register-trace normalization (ZS-M6/M44 v1.2); $C_{norm} = 1$ **DBI** at register tree level, **OPEN** at full parent 1PI; M_{UV} **PROVEN-irreducible** | (A, Q, dim Z) = (35/437, 11, 2) **LOCKED**.

§0. Abstract

ZS-F35 reduced the Z-Spin odd-sector vacuum susceptibility to

$\chi_-^{(s)} = (dim Y)^2, (A/Q), C_{norm}, M_{UV}^4 = (1260/4807), C_{norm}, M_{UV}^4, C_{norm} = \tilde{G}_s^{-1} (c_e/2\pi)^2$, and left a dimensionless gate ($\tilde{G}_s = 1$ ASSUMPTION, $c_e = 2\pi$ HYPOTHESIS-strong) and one dimensionful scale M_{UV} . Following the principle that only the field-redefinition invariant e_s^2/Z_s^{phys} is physical, this paper **organizes their normalization, topological quantization, and residual obstruction within a common parent-action template** — the honest current state is a *template*, not a completed parent-action computation ($\tilde{G}_s$ conditional on register-trace normalization, c_e on $\alpha_{UV} = 1$, C_{UV} OPEN at full 1PI, M_K an independent scale).

(NG1) [PROVEN] Under $A_3 \rightarrow \lambda A_3$: $Z_s \rightarrow Z_s/\lambda^2$, $e_s \rightarrow e_s/\lambda$; the invariant is e_s^2/Z_s , and C_{norm} is invariant, so $\tilde{G}_s = 1$ and $c_e = 2\pi$ are **not independent** — two projections of one convention.

(T1) [PROVEN] On $Y_6 = M_4 \times \Sigma_2$: $Z_s^{phys} = K_\Sigma/g_6^2$; for the unit-period form $K_\Sigma = A_\Sigma^{-1}$ (**inverse area** — correcting the V_Σ symbol). $\tilde{G}_s = 1 \Leftrightarrow g_6^2 = g_{reg}^2 K_\Sigma$.

(Cycle + T2/T3) [PROVEN / DERIVED-CONDITIONAL] The internal cycle is the **corpus-forced Koenigs torus** $\Sigma_2 = E_\lambda^\bullet$ (F34: the canonical rank-one homology generator, $b_2 = 1$; unique up to homology). $E_\lambda^\bullet = T^2$ is parallelizable, so $w_1 = w_2 = 0$ (spin), $W_3 = 0$ (Freed–Witten), $Tor, H^\bullet = 0$, $p_1(T^2) = 0$: **no intrinsic shift from the internal tangent bundle**. The full flux-quantization shift is **DERIVED-CONDITIONAL** on the vanishing of the relevant *ambient* shifted classes $(\lambda_{grav}(Y_6)|_{W_5} = 0$, $[H]|_{W_5} = 0$, $W_3(NW_5) = 0$), which the internal $p_1(T^2) = 0$ does not by itself establish. The wrapped-brane charge pairing has Smith normal form [1] (primitive), so the minimal charge is one unit and the *dimensionless* WZ phase is exactly 2π (**PROVEN**). The *dimensionful* $c_e = 2\pi$ then holds **iff** $\alpha_{UV} = 1$ (review 1.5); since $\alpha_{UV} = 1$ is not separately derived from the action, $c_e = 2\pi$ is **DERIVED-CONDITIONAL**.

(T4) [T4a PROVEN / T4b–T4c DERIVED-CONDITIONAL] Full-flux partition matching splits into: **T4a** theta-series rigidity — if $Z_{EFT}(\theta) = Z_{reg}(\theta)$ then $Z_s^{phys} = 1/g_{reg}^2$ (**PROVEN**, Fourier rigidity); **T4b** whether the register action *generates* that theta series (**DERIVED-CONDITIONAL** on register-trace normalization); **T4c** $\tilde{G}_s = 1$ (**DERIVED-CONDITIONAL**). The **algebraic back-definition is avoided** ($g_{reg}^2 = 6A/Q$ is upstream ZS-M6, not back-defined), **but the register-trace bridge remains conditional** — per ZS-M44 v1.5, $\rho_Q = I_Q/Q$ is asserted, not action-derived (§6).

(T5) [DBI at tree level / OPEN at full 1PI] Threshold audit $Z_s^{phys} = Z_{tree} + \Delta_{bulk} + \Delta_{brane} + \Delta_{ct}$: at the **register tree level**, $\Delta_{bulk} = 0$ (the Z_2 -odd slot is decoupled, ZS-M6 §10.3) and the register branch is DBI. **But (review 1.6) the full parent 1PI is OPEN**: $F_4 \wedge \star F_4$ is gauge-invariant and seam-even, so gauge invariance + seam parity do **not** forbid the Δ_{ct} counterterm, and the absence of a propagating 4D 3-form mode does not prevent heavy parent fields from renormalizing the F_4^2 coefficient. So: register tree branch DBI; full parent 1PI **OPEN**; instanton residual $\Delta_{brane} = O(e^{-S_{inst}})$ **HYPOTHESIS/EXPECTED**.

(Collapse) [DERIVED, with caveat] $\chi_-^{(s)} = \tilde{g}^2$; adopting $C_{norm} = 1$ is the *definition*

$M_{UV}^4 := (4807/1260)\tilde{g}^2$ (a canonical reparameterization, review 1.7), so $\chi_-^{(s)} = (1260/4807)M_{eff}^4$ with $M_{eff} := C_{UV}^{1/4}M_K$ is an **IDENTITY / canonical reparameterization** (not a DERIVED prediction); C_{UV} (true 1PI matching) and M_K (metric scale) remain the physical content.

(T6/T7) [PROVEN] The Koenigs *shape* (modulus $\tau = (\omega + i\kappa_\lambda)/2\pi$) is M1-fixed, but the physical area $A_\Sigma = \ell^2$, Im, τ needs the linearizing length $\ell = M_{UV}^{-1}$; radion stabilization ($V_{eff} = c_{flux}/a + c_{brane}a$, minimum $a_* = \sqrt{c_{flux}/c_{brane}}$) relocates M_{UV} onto the parent scale M_* . By Buckingham- π no

dimensionless datum fixes a scale — the F-sector instance of the ZS-A27 A–Q–Only No-Go, consistent with ZS-A28 / ZS-F33.

Result (honest terminus). $\chi_-^{(s)} = \frac{1260}{4807}, C_{UV}, M_K^4$ where $1260/4807 = (dimY)^2 A/Q$ is the **register structural factor** (DERIVED-CONDITIONAL on register-trace normalization, ZS-M44 v1.5); the primitive-charge lattice and Koenigs complex structure are **closed**; C_{UV} is **conditional on full parent 1PI matching** (OPEN, review 1.6); M_K is the **independent Kähler/metric scale**; and M_K/M_P is a **genuine OPEN** (the metric-scale gate). Setting $C_{UV} = 1$ and $M_K = M_{UV}$ recovers the canonical form. **ZS-A31** consolidates the scale: $\rho_{\Lambda Z}/M_{eff}^4 = \frac{1}{2}(1260/4807)\omega^2$ (a definitional identity, since $M_{eff} := C_{UV}^{1/4} M_K$), regression fixing $M_{eff} \approx 2.48$ meV; the hierarchy $M_K/M_P^- = e^{-2\pi Q}$ is OPEN (the 2π Borchers–Wiesbrock-forced modular-depth frontier, ZS-M46/M47). Verification 56/56 + 14/14 PASS.

Epistemic Status Legend

Tag	Meaning
PROVEN	Explicit proof or exact machine verification; no undischarged assumption.
IMPORTED-PROVEN	Proven in external literature and used without re-proof; cited.
DERIVED	Follows from PROVEN results by stated steps; no new parameter.
DERIVED-BY-INHERITANCE	Uses an upstream corpus result (here ZS-M6), not re-proven; inherits its caveats.
DERIVED-CONDITIONAL	Derived modulo explicitly named, falsifiable conditions.
COMPUTED	A finite decidable computation carried out; result reported as-is.
HYPOTHESIS-strong	Structurally motivated; a key value not yet proven.
NO-GO	A proven impossibility/non-uniqueness disciplining the program.
PROVEN-irreducible	Proven that the quantity cannot be fixed by the stated inputs.
CLOSED-NEGATIVE	A route proven not to work (a no-go), possibly scope-limited.
IDENTITY / REGRESSION	An arithmetic or cross-version consistency; no new physical content.
NON-CLAIM	Explicitly outside the scope of this paper.
OPEN	Conceptually unresolved; a genuine gap.

§1. Introduction: the inherited ZS-F35 frontier and the parent-action mandate

ZS-F33 gave $\rho_{\Lambda Z} = \frac{1}{2}\chi_- \omega^2$ and the **Charge-Unit Obstruction** ($\chi_- = e_-^2/(4\pi^2 Z_-)$, unit not fixed by flux integrality). ZS-F34 isolated the unit under $Y_6 = M_4 \times \Sigma_2$ and identified the rank-one homology generator as the **Koenigs torus** E_λ^* ($\dim Z = 2$, $b_2 = 1$, unique up to homology; F33.2B PROVEN). ZS-F35 collapsed the normalizations into $\tilde{G}_s (Z_s^{phys} = \tilde{G}_s/g_{reg}^2)$, proved $\hat{c}_\theta = \star$, $v_s^2 = 6 = dimY$, and reached

$$\chi_-^{(s)} = (dimY)^2, \frac{A}{Q}, C_{norm}, M_{UV}^4 = \frac{1260}{4807}, C_{norm}, M_{UV}^4, \quad C_{norm} = \tilde{G}_s^{-1} (c_e/2\pi)^2.$$

Mandate (F36). Do not declare $\tilde{G}_s = 1$, $c_e = 2\pi$, M_{UV} separately; by NG1 only e_s^2/Z_s^{phys} is physical. Organize it within a common parent-action template, on the corpus-forced cycle, with every normalization traced to an upstream corpus result, flagged conditional, or proven irreducible. v1.2 completes this: the cycle is E_λ^* (not a free ansatz); the charge and anomaly structure are *computed* on it; the register normalization is *inherited* from ZS-M6; and the residual scale is proven irreducible and localized to the loxodromic linearizing length.

Locked data: $A = 35/437$, $Q = 11$, $(Z, X, Y) = (2, 3, 6)$, $\kappa^2 = A/Q = 35/4807$,
 $g_{reg}^2 = \dim(Y)\kappa^2 = 210/4807$, $\omega = 2.2592495540$, $\kappa_\lambda = -\ln|\lambda^*| = 0.1148346250$,
 $z^* = 0.43828 + 0.36059i$.

§2. The Normalization-Covariance Theorem (F36.NG1)

$$S_4 = -\frac{Z_s}{2} \int_{M_4} F_4 \wedge \star F_4 + e_s \int_{W_3} A_3, \quad F_4 = dA_3.$$

Theorem F36.NG1. [PROVEN] Under $A_3 \rightarrow \lambda A_3$: $Z_s \rightarrow Z_s/\lambda^2$, $e_s \rightarrow e_s/\lambda$; e_s^2/Z_s is invariant,

$\chi_-^{(s)} = e_s^2 / (4\pi^2 Z_s^{phys})$ is the unique odd-sector field-redefinition invariant, and $C_{norm} = \tilde{G}_s^{-1} (c_e/2\pi)^2$ is invariant.

Proof. Substituting $A_3 = A_3'/\lambda$ gives the transformations; then $e_s'^2/Z_s' = e_s^2/Z_s$, and with $\tilde{G}_s = Z_s^{phys} g_{reg}^2$,
 $c_e = e_s / (\nu_s M_{UV}^2)$, $C_{norm} = \chi_-^{(s)} / (g_{reg}^2 \nu_s^2 M_{UV}^4)$.

Exhibiting $\tilde{G}_s = 1$ and $c_e = 2\pi$ separately does *not* exclude a compensating field redefinition; the only meaningful target is C_{norm} , computed in one fixed normalization (§§3–7). The split of $C_{norm} M_{UV}^4$ into “ C_{norm} ” and “ M_{UV} ” is a convention (the seed of §8). (*Verified: T1 e_s^2/Z_s invariant, T1 C_{norm} field-redef invariant.*)

§3. The compact parent action and the exact reduction (F36.T1)

$$Y_6 = M_4 \times \Sigma_2, \quad C_5 = A_3^- \wedge \omega_2, \quad G_6 = dC_5 = F_4^- \wedge \omega_2, \quad \int_{\Sigma_2} \omega_2 = 1,$$

$$S_6 = -\frac{1}{2g_6^2 Y_6} \int G_6 \wedge \star_6 G_6; +; 2\pi i N \int_{W_5} C_5; +; S_{metric} + S_{stab} + S_{ct},$$

C_5 a differential-cohomology character [Hopkins–Singer], $W_5 = W_3 \times \Sigma_2$. **Scope caveat (ZS-F33):** the “6” is a rank-2 internal fibre over the 4D base, not fundamental 6D spacetime.

Theorem F36.T1 (Exact reduction). [PROVEN] *For the product metric, $\star_6(F_4 \wedge \omega_2) = (\star_4 F_4) \wedge (\star_2 \omega_2)$, the integral factorizes, and*

$$Z_s^{phys} = \frac{K_\Sigma}{g_6^2}, \quad K_\Sigma := \int_{\Sigma_2} \omega_2 \wedge \star_2 \omega_2 = A_\Sigma^{-1} \text{ (inverse area)}, \quad \tilde{G}_s = 1 \Leftrightarrow g_6^2 = g_{reg}^2 K_\Sigma.$$

With $\omega_2 = vol_\Sigma / A_\Sigma$ one keeps $A_\Sigma = \int vol_2$ (area), $\int \omega_2 = 1$ (period), $K_\Sigma = A_\Sigma^{-1}$ (kinetic factor) **distinct**; the ZS-F33/F34 V_Σ is $K_\Sigma = A_\Sigma^{-1}$, an *inverse* area — conflation would invert the M_{UV} power. (*Verified on S_R^2 ; the same identity holds on any connected oriented Σ_2 , in particular E_λ : T2a, T2b, T2b'.*)

§4. Wrapped-brane charge and the 2π (F36.T2)

$$\exp\left(2\pi i N \int_{W_5} C_5\right), \quad C_5 = A_3 \cup \omega_2, \quad \int_{\Sigma_2} \omega_2 = 1; \Rightarrow; 2\pi i N \int_{W_3} A_3.$$

The minimal charge coefficient of the dimensionless integral field is 2π (large-gauge / generalized Dirac).

Writing $A_3 = \alpha_{UV} M_{UV}^2 a_3$; $e_6 = (2\pi/\alpha_{UV}) M_{UV}^2$, so

$$, c_e = 2\pi \Leftrightarrow \alpha_{UV} = 1,,$$

the **same** normalization as $\tilde{G}_s = 1$. That the minimal charge is exactly one unit (hence coefficient 2π , not a fraction) is *proven* by the charge pairing of §5. (*Verified: T2-NS $c_e=2\pi \Leftrightarrow \alpha_{UV}=1$.*)

§5. The Koenigs cycle $\Sigma_2 = E_\lambda^*$: charge pairing and anomaly gates (F36.T3)

The internal cycle is not a free choice: ZS-F34 proves the rank-one generator of H_2 carrying the bivector 2-form is the Z-sector **Koenigs torus** $E_\lambda^* = C^* / \langle \lambda^* \rangle$ ($\dim Z = 2, b_2 = 1$; F33.2B). We compute the charge and anomaly structure on it. As a smooth manifold $E_\lambda^* = T^2$, which is **parallelizable**.

Charge pairing (Smith normal form). [PROVEN] The cellular chain complex of T^2 (one vertex, two edges, one face, all boundary maps zero) gives Betti $(1, 2, 1)$, torsion-free, with $H_2(T^2, Z) = Z = \langle [\Sigma_2] \rangle$.

The wrapped-brane flux pairing $\langle [\Sigma_2], \omega_2 \rangle = \int_{\Sigma_2} \omega_2 = 1$ has Smith normal form [1] — **primitive** — so

the minimal wrapped charge is $q_{min} = 1$ (**PROVEN**) and the *dimensionless* WZ phase $S_{WZ} = 2\pi i \int A_3$ is exactly 2π (**PROVEN in integral normalization**). The *dimensionful* $c_e = 2\pi$ then holds **iff** $\alpha_{UV} = 1$ —

DERIVED-CONDITIONAL (review error 3). The H_1 intersection form is the unimodular symplectic $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ (self-dual lattice). (*Verified: M2a–M2e.*)

Anomaly gates. [DERIVED-CONDITIONAL on ambient and normal-bundle classes]. Because T^2 is parallelizable, $w(TT^2) = 1$, so:

- **A1 (bulk shift):** $p_1(T^2) = 0$ (flat) removes the *intrinsic* internal-tangent contribution only; full unshifted flux quantization additionally requires the ambient class $\lambda_{grav}(Y_6)|_{W_5} = 0$, on which $\text{flux} \in 2\pi\mathbb{Z}$. (*Verified: M3e.*)
- **A2 (Freed–Witten):** $w_2 = 0$ (spin) $\Rightarrow W_3 = \beta(w_2) = 0$; with trivial normal bundle and no worldvolume H -flux, $W_3(\text{normal}) + [H]|_{W_5} = 0$ — anomaly-free, no worldvolume gauge field forced. (*Verified: M3a–M3c.*)
- **A3 (torsion):** $Tor, H^\bullet(T^2, \mathbb{Z}) = 0 \Rightarrow$ trivial linking pairing, no charge shift. (*Verified: M3d.*)

Upgrade over v1.1. In v1.1 these gates were conditional on a *free* S^2 ansatz. Now $\Sigma_2 = E_\lambda^*$ is corpus-forced and the classes are computed exactly on it; the only residual conditions are the *ambient* and *normal-bundle* shifted classes ($\lambda_{grav}(Y_6)|_{W_5} = 0$, $[H]|_{W_5} = 0$, $W_3(NW_5) = 0$) — genuine physical conditions on the brane embedding, not an ansatz choice; the internal $p_1(T^2) = 0$ discharges only the intrinsic internal-tangent piece.

§6. $\tilde{G}_s = 1$ by full-flux partition matching: theta-rigidity PROVEN, physical matching conditional (F36.T4)

On compact Euclidean M_4 ($\int_{M_4} F_4 = 2\pi n$), $S_n = 2\pi^2 Z_s^{phys} n^2 / V_4$, so

$$Z_{EFT}(\theta) = \sum_n e^{-2\pi^2 Z_s^{phys} n^2 / V_4 + in\theta}, \quad Z_{reg}(\theta) = \sum_n e^{-2\pi^2 n^2 / (g_{reg}^2 V_4) + in\theta}.$$

Theorem F36.T4 (split into three honest tiers).

- **T4a (theta-series rigidity) [PROVEN].** *If the two theta series are equal, $Z_{EFT}(\theta) = Z_{reg}(\theta)$ for all (n, θ, V_4) , then $Z_s^{phys} = 1/g_{reg}^2$.* This is pure Fourier rigidity (below).
- **T4b (physical partition matching) [DERIVED-CONDITIONAL].** *Does the register action actually generate the EFT theta series?* This is the register-trace normalization bridge —

conditional on $\rho_Q = I_Q/Q$ (ZS-M44 v1.3 §3/§6).

- **T4c (normalization) [DERIVED-CONDITIONAL].** $\tilde{G}_s = 1$, conditional on T4b and register-trace normalization.*

Proof. (T4a only.) Both are theta series $\sum_n e^{-an^2+in\theta}$; the θ -Fourier coefficient at mode n is e^{-an^2} , so equality for all θ needs $a = b$ already at $n = 1$ — **no rescaling freedom**. This rigidity is PROVEN. Whether the register action *produces* this series (T4b) rests on the register-trace normalization, so $\tilde{G}_s = 1$ (T4c) is **DERIVED-CONDITIONAL**. ; ◻ (*Verified: T4a–T4d.*)

Algebraic back-definition avoided; register-trace bridge conditional. The matching avoids circular back-definition only if g_{reg}^2 is an *independent* upstream quantity. It is: **ZS-M6 Theorem 2.2.1**

(Register-Total Normalization) derives $\kappa^2 = A/Q$ from the 10-step chain C1–C10 (non-minimal coupling, Regge discretization, Mode-Count Collapse, Gilkey heat kernel, spectral asymmetry, Peter–Weyl/Schur), and **ZS-M6 Theorem 2.2.2 (Dimensional Coupling Norm)** gives $g_\Gamma^2 = \dim(\Gamma)\kappa^2$, whence

$$g_{reg}^2 = \dim(Y) \cdot \kappa^2 = 6, \frac{A}{Q} = \frac{210}{4807}, \quad \frac{1}{g_{reg}^2} = \frac{4807}{210}.$$

This is a *projection* from the register Block-Laplacian, provenance {Peter–Weyl, Schur, rank-1 β_0 -selection} — independent of the F35 target. **Status after review (ZS-M44 v1.5, integrating M45):** the firm advance is the R-2 perturbation-artifact correction (the $0.855 \approx \delta_x/\delta_y$ trap defused; democratic within irreps via Schur). **But the register-scalar normalization is not fully discharged:** (a) the “/Q” is the normalized-trace rank $a_0 = \dim = Q$ — *not* $a_1 = Tr, L$ (the earlier “ a_1 mode count” was an error); (b) the maximally-mixed density $\rho = I_Q/Q$ underlying the democratic distribution is *asserted*, not action-derived; (c) ZS-M44 v1.5 §6.3 provides a graph-Laplacian **coordinate diagnostic** (Rayleigh quotients 0.600 vs 0.244; these are *not* eigenvalues, and *not* the genuine $B^\top W_{hinge} B$ Regge Hessian, which remains OPEN) — evidence the sector diagonals differ, narrowing the residual to the action-level selection of a register measure (a_0 mode-count vs metric being two benchmarks). Hence $\tilde{G}_s = 1$ is **DERIVED-CONDITIONAL on register-trace normalization**. This still upgrades ZS-F35’s S3 from ASSUMPTION, and is **scale-free** (M_{uv} never appears). (*Verified: M1a–M1d; ZS-M44 v1.5 19/19 + 14/14 + 12/12.*)

§7. The 1PI threshold audit, with bulk-vanishing grounded in ZS-M6 (F36.T5)

$$Z_s^{phys}(\mu) = K_\Sigma / g_6^2 + \Delta_{bulk}(\mu) + \Delta_{brane}(\mu) + \Delta_{ct}(\mu).$$

$\sim Z_{tree}$

Theorem F36.T5. [register tree DBI / full 1PI OPEN]

- $\Delta_{ct} = 0$ [**DERIVED-CONDITIONAL, review 1.6**]. At the register tree level the 4D 3-form has no local propagating bulk degree ($F_4 = \star f$, f constant). **However, this does not close the full 1PI:** $F_4 \wedge \star F_4$ is gauge-invariant and seam-even, so gauge invariance + seam parity alone do **not** forbid a Δ_{ct} counterterm, and heavy parent fields may renormalize the F_4^2 coefficient even absent a propagating 3-form. So $\Delta_{ct} = 0$ holds for the register tree branch (DBI); the full parent 1PI is **OPEN**.
- $\Delta_{bulk} = 0$ [**DERIVED-BY-INHERITANCE**]. In the ZS-M6 register Block-Laplacian, the Z_2 **-odd** mode (slot 1 — the odd singlet) has eigenvalue shift $\Delta\lambda = 0$ under the rank-1 β_0 -selected cross-coupling: it is **completely decoupled** (ZS-M6 §10.3, hand-proven; matches the ZS-S1 §5.2 Z_2 -projection). This is the register shadow of the parent statement that no charged mode couples locally to the odd singlet at $p = 0$; the full parent statement carries a residual **DERIVED-CONDITIONAL** on the (unspecified) heavy odd content. (*Verified: M6a, M6b.*)
- $\Delta_{brane} = o(e^{-S_{inst}})$ [**HYPOTHESIS/EXPECTED**]. Membrane-nucleation corrections $\propto e^{-T_{mem} A_\Sigma}$ are *expected* to be exponentially small, but the instanton solution and its fluctuation determinant are **not computed** here — so this is **EXPECTED**, not **COMPUTED**.

Under the canonical branch (ZS-F33.6, $b_{2n} = 0$), the **register tree branch** gives $Z_s^{phys} = Z_{tree}$ and

$$C_{norm}^{reg,tree} = 1 \text{ [DBI]}, \quad C_{UV}^{1PI} = 1 + \Delta_{bulk} + \Delta_{ct} + \Delta_{brane} \text{ [OPEN]}.$$

So $C_{norm} = 1$ holds **only for the register tree branch (DBI)**; the **full parent 1PI C_{UV} is OPEN** (Δ_{ct} not forbidden, Δ_{brane} EXPECTED not computed). Any gate failure would be reported as a *computed* $C_{UV} \neq 1$ — which would be a *more* valuable result than matching 1 (review §2.3). (*Verified: T5a, T5b, T5c.*)

§8. The residual collapse: $\chi_-^{(s)} = \tilde{g}^2$

With $\tilde{G}_s = 1$ (§6) and $c_e = 2\pi$ (§4–5) in one normalization, write $A_3 = c, a_3$ (a_3 the integral field, coupling $\tilde{g}$): $Z_s^{phys} c^2 = 1/\tilde{g}^2$, $e_s c = 2\pi$, so

$$\chi_{-}^{(s)} = \frac{e_s^2}{4\pi^2 Z_s^{phys}} = \frac{(2\pi/c)^2}{4\pi^2 (1/\tilde{g}^2 c^2)} = \tilde{g}^2,$$

the conversion c (hence $\tilde{G}_s, c_e$ separately) cancels. Cross-check with (1.1): $\chi_{-}^{(s)} = v_s^2 g_{reg}^2 C_{norm} M_{UV}^4 = \tilde{g}^2$.

Defining the **effective scale** $M_{eff} := C_{UV}^{1/4} M_K$ gives the canonical *reparameterization*

$M_{eff}^4 := (4807/1260)\tilde{g}^2$ — the definition of M_{eff} does **not** require adopting $C_{UV} = 1$. **The convention-dependent split $(\tilde{G}_s, c_e)$ collapses to the invariant $\tilde{g}^2$, but the physical parameterization retains two residuals: C_{UV} (OPEN, full 1PI) and M_K (independent scale). (Verified: $RC\ chi=\tilde{g}^2$, $RC\ C_{norm}\ c$ -independent.) [DERIVED as a reparameterization identity; the physical content C_{UV}, M_K is OPEN.]**

§9. The Koenigs modulus, radion stabilization, and the sharpened scale gate (F36.T6)

Shape is fixed by M1; scale is not. [PROVEN] The Koenigs torus $E_{\lambda^*} = C/(2\pi iZ + (\log \lambda^*)Z)$ has modulus

$$\tau = \frac{\log \lambda^*}{2\pi i} = \frac{\omega + i\kappa_{\lambda}}{2\pi}, \quad Im, \tau = \frac{\kappa_{\lambda}}{2\pi} > 0,$$

fixed by the loxodromic multiplier $\lambda^* = f'(z^*)$ of ZS-M1 ($|\lambda^*| = 0.89151, \kappa_{\lambda} = 0.11483, \omega = 2.2592$). Thus the **shape** and the **dimensionless** fundamental-domain area are M1-determined. But the **physical** area is $A_{\Sigma} = \ell^2, Im, \tau$, where ℓ is the physical size of the loxodromic linearizing coordinate — an undetermined length, $\ell = M_{UV}^{-1}$. Topology + M1 fix everything *except* ℓ . (Verified: M4a–M4d.)

Theorem F36.T6 (Radion relocation). [DERIVED-CONDITIONAL] *On T^2 the curvature term vanishes by Gauss–Bonnet ($\chi = 0$), so stabilization is flux-versus-tension:*

$$V_{eff}(a) = \frac{c_{flux}}{a} + c_{brane}, a, \quad a_* = \sqrt{c_{flux}/c_{brane}}, \quad V_{eff}''(a_*) > 0$$

(a genuine minimum, no tachyonic radion). But $c_{flux} \sim g_6^{-2} n^2$ and $c_{brane} \sim T_4$ are parent scales, so $M_{UV} = M_*$, $h(\text{couplings})$: stabilization relocates the scale onto M_* , it does not create one. (Verified: M5a–M5d.) If T_4/M_* is external, M_{UV} is DERIVED-CONDITIONAL, never DERIVED-from-(A, Q).

§10. The F-sector Scale No-Go, final susceptibility, and firewall (F36.T7)

Theorem F36.T7 (F-sector Scale No-Go). [PROVEN] $\tilde{g} = \chi_-^{(s)}$ (equivalently M_{UV}) is not a function of the locked dimensionless data $(A, Q, \dim Z, v_s^2, \omega, \kappa_\lambda, z^*, \dots)$ + flux integrality + differential-cohomology normalization alone.

Proof. By NG1 the physical content is the invariant $\chi_-^{(s)}$ of dimension $[mass^4]$; by T1–T5 every dimensionless input is a pure number and the topology contributes only integers and 2π . Buckingham- π forbids a dimensionful function of dimensionless inputs. The §9 Koenigs analysis exhibits the missing datum **concretely**: the physical size ℓ of the loxodromic linearizing coordinate — invisible to cohomology (which fixes $b_2 = 1$) and to M1 (which fixes the shape τ). ; (Verified: M4d, M5c, M5d.)

This is the odd-three-form realization of the **ZS-A27 A–Q-Only Scale-Generation No-Go**, consistent with **ZS-A28** (internal scale generation closed-negative) and **ZS-F33** (Charge-Unit Obstruction), and *strengthens* F33: the full differential-cohomology parent normalization + partition matching + threshold audit + Koenigs modulus still cannot fix the scale.

Routes to $f = M_{UV}/M_P^-$ (non-exhaustive, ZS-A27 §7). With the one anchor M_P^- : (i) target-fit exponent

— NUMEROLOGY, barred; (ii) transmutation $\Lambda_- = M_P^- e^{-8\pi^2/b_0 g_-^2}$,

$b_0 = \frac{11}{3}C_2(G_-) - \frac{2}{3}\sum_f T(R_f) - \frac{1}{6}\sum_s T(R_s)$ — LEGITIMATE only if G_- and $g_-^2(\mu_0) = g_{reg}^2$ are

corpus-forced; ZS-F33 leaves G_- unspecified $\Rightarrow$ **CLOSED-NEGATIVE-within-corpus**; (iii) $\rho \sim \Lambda^3 H$

— dynamical DE ($w \neq -1$), contradicting ZS-F33.7/A28 $\Rightarrow$ **NON-CLAIM**. Within the corpus, M_{UV} is **PROVEN-irreducible**.

Final susceptibility — physical general form and canonical reparameterization (IDENTITY):

$\chi_-^{(s)} = \frac{1260}{4807}, C_{UV}, M_K^4$; (physical general form; C_{UV} OPEN at full 1PI, M_K independent Kähler scale),

$M_{eff} := C_{UV}^{1/4} M_K, \quad \chi_-^{(s)} = \frac{1260}{4807}, M_{eff}^4 \approx 0.2621, M_{eff}^4$,
 $\sim$ IDENTITY / canonical reparameterization — not a DERIVED prediction

The second line is a **definitional identity**, not an extra prediction; calling M_{eff} “ M_{UV} ” is permitted, but then the ZS-A31 one-parameter reduction is an *effective parameterization*, not a physical closure.

$w = -1$ is **DERIVED-BY-INHERITANCE within the frozen/canonical branch** (ZS-F33.7/A28) — not an observationally proven theorem. The frozen branch remains observationally viable, although **DESI DR2 combinations show dataset-dependent preferences for evolving dark energy** relative to flat Λ CDM (BAO+CMB $\sim 3.1\sigma$; adding SNe $\sim 2.8\text{--}4.2\sigma$ depending on the sample), so this is a genuine falsification channel, not merely “not excluded.”

Anti-numerology firewall. Code split derivation/{parent_action, charge_pairing, stiffness, partition_match, threshold, stabilization} $\perp$ regression/{observed_rho_lambda}; loading observation

before derivation fails the gate (M7a, M7b). Post-firewall, matching (10.1) to the inferred dark-energy density $\rho_\Lambda^{1/4} \approx 2.3 \text{ meV}$ fixes $M_{eff} \approx 2.5 \text{ meV}$ — a **REGRESSION on one input, not a prediction**.

The length $\ell = M_K^{-1} \approx 80, \mu\text{m}$ follows **only** on the canonical register-tree branch $C_{UV} = 1$ (there $M_K = M_{eff}$), and $M_K/M_P^- \approx 10^{-30}$ is then the metric-scale gate. Cosmological consolidation $\rightarrow$ **ZS-A31**.

§11. Claim ledger

#	Claim	Status	Confidence	Conditions
NG1	$e_s^2/Z_s, c_{norm}$ field-redef invariant; $\tilde{G}_s, c_e$ one convention	PROVEN	—	none
T1	$Z_s^{phys} = K_\Sigma/g_6^2$; $K_\Sigma = A_\Sigma^{-1}$	PROVEN	—	product metric
Cyc	$\Sigma_2 = E_\lambda^*$ (rank-one H_2 generator, $b_2 = 1$)	DERIVED-BY-IN HERITANCE	—	F34 / F33.2B
CP	Smith form [1], $q_{min} = 1$, dimensionless WZ phase 2π	PROVEN	—	none
CP'	physical $c_e = 2\pi$	DERIVED-CON DITIONAL	—	$\alpha_{UV} = 1$
T3	anomaly gates ($W_3 = 0$, torsion $= 0, p_1 = 0$) on E_λ^*	DERIVED-CON D	80%	$\lambda_{grav}(Y_6) _{W_5} = 0, [H] _{W_5} = 0,$ $W_3(NW_5) = 0$
T2-NS	dimensionless WZ phase 2π PROVEN; $c_e = 2\pi \Leftrightarrow \alpha_{UV} =$	DERIVED-CON DITIONAL	—	$\alpha_{UV} = 1$
T4a	theta-series rigidity ($Z_{EFT} = Z_{reg} \Rightarrow Z_s^{phy}$)	PROVEN	—	Fourier rigidity
T4b/c	register generates the series; $\tilde{G}_s = 1$	DERIVED-CON DITIONAL	72%	register-trace normalization (ZS-M44 v1.5)
T5	$C_{norm}^{reg,tree} = 1$ DBI; C_{UV}^{1PI} OPEN ; Δ_{brane} EXPECTED not computed	DBI / OPEN	55%	full parent 1PI matching

#	Claim	Status	Confidence	Conditions
Coll	$\chi_-^{(s)} = \tilde{g}^2$; convention split ($\tilde{G}_s, c_e$) collapses to invariant $\tilde{g}^2$, but (C_{UV}, M_K) remain	DERIVED	90%	NG1–T5
Canon	$\chi_-^{(s)} = (1260/48)$, $M_{eff} := C_{UV}^{1/4} M_K$	IDENTITY / REPARAMETER IZATION	—	definitional
T6	radion minimum $a_* = \sqrt{c_{flux}/c_{brane}}$; relocation	DERIVED-CON D	65%	stable minimum; parent scales
Mod	Koenigs shape τ M1-fixed; physical ℓ free	PROVEN	—	none
T7	M_{UV} not fixed by dimensionless data + topology	PROVEN-irreduci ble	90%	Buckingham- π + §9
§10	$M_{eff} \approx 2.5$ meV	REGRESSION	—	firewalled, not a prediction
§10	$\ell = M_K^{-1} \approx 80$, m	DERIVED-CON DITIONAL	—	$C_{UV} = 1$ (canonical register-tree branch)

§12. Falsification gates

- **F-F36.1.** If $A_3 \rightarrow \lambda A_3$ fails to give $Z_s \rightarrow Z_s/\lambda^2$, $e_s \rightarrow e_s/\lambda$, NG1 fails. *PASS*.
- **F-F36.2.** If $K_\Sigma \neq A_\Sigma^{-1}$ or the reduction sign is wrong, T1 fails. *PASS (symbolic)*.
- **F-F36.3.** If the E_λ charge pairing were non-primitive (Smith form $\neq [1]$), or if $W_3 \neq 0$ / torsion $\neq 0$ on the forced cycle, $c_e \neq 2\pi$. *PASS (computed)*.
- **F-F36.4.** If ZS-M6's $g_{reg}^2 = \dim(Y)\kappa^2$ were back-defined rather than projected from the Block-Laplacian, T4 is circular. *Guarded by ZS-M6 C1–C10 provenance; R-1/R-2/R-3 flagged*.
- **F-F36.5.** If a charged bulk mode coupled to the odd singlet at $p = 0$ (contra ZS-M6 §10.3), or a kinetic counterterm were symmetry-allowed, or an instanton unsuppressed, $C_{norm} \neq 1$.

- **F-F36.6.** A dimensionful M_{UV} as a function of $(A, Q) + \text{integers}$ alone would violate Buckingham- π — impossible; any such claim is numerology.
 - **F-F36.7.** A computed ρ_Λ inconsistent with observation once M_{UV} is fixed by an *independent* UV input falsifies the identification (no retuning). $w \neq -1$ at high significance forces the dynamical branch (NON-CLAIM here).
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§13. Anti-numerology

Claimed numbers: (i) $1260/4807 = (\dim Y)^2 A/Q$, locked, inherited PROVEN; (ii) $v_s^2 = 6$ (ZS-F35 T1); (iii) $\kappa^2 = A/Q$, $g_{reg}^2 = 6\kappa^2$ (ZS-M6, DERIVED upstream); (iv) $c_e = 2\pi$ (charge-pairing Smith form, PROVEN); (v) $\tilde{G}_s = 1$ (partition matching + ZS-M6); (vi) $C_{norm} = 1$ (a normalization = the definition of M_{UV}). **No coefficient fit to data.** M_{UV} is explicitly not computed from (A, Q) ; its irreducibility is PROVEN, and the missing datum is localized to the loxodromic length ℓ . The $M_{eff} \approx 2.5$ meV value is loaded only behind the derivation/ $\perp$ regression/ firewall. The pre-registered gate (no fitted exponent / half-weight / base-10 coincidence / back-solved scale) passes by construction.

§14. Cross-version safety

- $(A, Q, \dim Z) = (35/437, 11, 2)$, $g_{reg}^2 = 210/4807$, $\kappa^2 = 35/4807$, $\omega = 2.2592495540$, $\kappa_\lambda = 0.1148346250$, $z^* = 0.43828 + 0.36059i$: **unchanged**.
- **ZS-M1:** used only via (a) the exponential-homomorphism uniqueness $\sigma = e^x$ (flux gluing, F33.4B-ii) and (b) the loxodromic multiplier $\lambda^* = f'(z^*)$ fixing the Koenigs modulus τ (§9). $|f'(z^*)| = |\ln i| |z^*| = 0.89151 < 1$ (attracting), matching $\$^\wedge\$$. *Fixed point untouched.* **M1 $\rightarrow$ S1 $\rightarrow$ U1 chain safe.** (Verified: M1, M1b, M4a, M4b.)*
- **ZS-M6:** newly load-bearing. $g_{reg}^2 = \dim(Y)\kappa^2$ (Thm 2.2.1/2.2.2) discharges the §6 non-circularity; the odd-mode decoupling (§10.3) grounds the §7 bulk-vanishing. No ZS-M6 result is modified.
- **ZS-M44 (v1.5, integrates ZS-M45):** corrects the a_0/a_1 conflation ($Q = a_0 = \dim$, *not* $a_1 = Tr, L$); the firm advance is the R-2 perturbation-artifact correction (§10.4 trap defused, democratic within irreps via Schur). The register-scalar density $\rho = I_Q/Q$ is asserted, not

action-derived, so $\kappa^2 = A/Q$ and thus $\tilde{G}_s = 1$ are **DERIVED-CONDITIONAL on register-trace normalization** (§6).

- **ZS-M45 (integrated into ZS-M44 v1.5 §6):** cell-coefficient sector-independence ($\delta\varphi_{cell} = A$); the genuine register Regge Hessian $B^\top W_{hinge} B$ is uncomputed (§6.3 gives only a coordinate diagnostic). Operator equality **OPEN**.
- **ZS-A31 (v1.4):** consolidates the *effective* scale $M_{eff} = C_{UV}^{1/4} M_K$ (observation fixes $M_{eff} \approx 2.48$ meV only); the M_K/M_P hierarchy is OPEN, with the modular-depth $e^{-2\pi Q}$ frontier (2π Borchers–Wiesbrock-forced) the pre-registered candidate, scoped to ZS-M46/M47.
- **ZS-F33:** Charge-Unit Obstruction preserved and strengthened; frozen $w = -1$ and the canonical branch (F33.6) used. No status reversed.
- **ZS-F34:** master form unmodified; F36 supplies $\tilde{G}_s, c_e$ and the $K_\Sigma = A_\Sigma^{-1}$ correction; the $\Sigma_2 = E_\lambda$ identification is used verbatim. G-Outer-Physical inherited OPEN.
- **ZS-F35:** (1.1), $v_s^2 = 6$ used verbatim; S3 $\rightarrow$ DERIVED-BY-INHERITANCE, S4 $\rightarrow$ PROVEN; F-F35.5/F-F35.7 closed/sharpened.
- **ZS-A27 / A28:** T7 is the F-sector instance of the A27 A–Q-Only No-Go, consistent with A28. No cosmological status altered; scale handed to ZS-A31.

§15. Verification

zs_f36_verify_v2_1.py (SymPy; checks identical to v2.0 — the v2.1 edits are text-only), **56/56 PASS**, three blocks:

- **Core (17):** structural factor $36A/Q = 1260/4807$; $g_{reg}^2 = 210/4807$, $v_s^2 = 6$; NG1 invariances; T1 $K_\Sigma = A_\Sigma^{-1}$ on S_R^2 ; collapse $\chi_-^{(s)} = \tilde{g}$; M1 $|f'(z^*)| = 0.89151$; convergence $[5, 1, 0]$.
- **v1.1 (13):** T4 theta-matching (mode-1 forces $a = b$; $Z_s^{phys} = 4807/210$; full spectrum); T5 threshold; Σ_2 no-go $b_2(\Sigma_g) = 1$; $c_e = 2\pi \Leftrightarrow \alpha_{UV} = 1$.

- **Companion / parent-action (26):** register projection $g_{reg}^2 = \dim(Y)\kappa^2$ (M1a–d); charge pairing on T^2 — Betti (1, 2, 1), torsion-free, Smith form [1], unimodular intersection form (M2a–e); anomaly gates $w = 1$, $W_3 = 0$, torsion = 0, $p_1 = 0$ (M3a–e); Koenigs modulus τ M1-fixed, physical ℓ free (M4a–d); radion minimum $a_* = \sqrt{c_{flux}/c_{brane}}$, $V'' > 0$, relocation (M5a–d); odd-mode decoupling (M6a–b); firewall (M7a–b).

The verification is a **consistency + no-go certificate, not a closure certificate for M_{UV}** — by design. The six companion modules of the pre-registered architecture (f36_register_projection, f36_charge_pairing, f36_anomaly_gates, f36_koenigs_modulus, f36_radion_stabilization, f36_final_firewall) are now **executed**, not deferred.

§16. Acknowledgements & Code Availability

Consolidates internal Z-Spin Collaboration deep-exploration notes following ZS-F35 v1.5 and the pre-registered issue-tree architecture (Appendix A), including the parent-action module suite. `zs_f36_verify_v2_1.py` (SymPy) reproduces all 56 checks (identical to v2.0; the v2.1 edits are text-only). This work used AI tools (Anthropic Claude) for verification and drafting; the author assumes full responsibility, including the $K_\Sigma = A_\Sigma^{-1}$ correction, the $S^2 \rightarrow E_\lambda$ cycle correction, and the honest ZS-M6 and non-perturbative caveats.

§17. Appendix A — Deep-exploration (issue-tree) record

Step 0 (long list, 7). (1) normalization-covariance; (2) exact reduction + K_Σ ; (3) the 2π ; (4) $\tilde{G}_s = 1$ via partition matching / Hessian; (5) anomaly gates; (6) M_{UV} via stabilization/transmutation; (7) scale no-go.

Step 1 (MECE). I1 = (1); I2 = {(2)+(3)+(5)+(4-partition)+(4-threshold)} (all load-bearing); I3 = {(6) (7)}. Dropped: branch-ratio (tautological); dynamical- w (NON-CLAIM). *v1.2 parent-action sub-tree*: the cycle is forced to E_λ (not a free node); charge pairing, anomaly classes, register normalization, and odd-mode decoupling become *computed/inherited* nodes rather than assumptions.

Step 2–3 (tree + status). NG1 PROVEN; T1 PROVEN; Cyc DERIVED-BY-INHERITANCE (F34); $q_{min} = 1$ + dimensionless 2π PROVEN, dimensionful $c_e = 2\pi$ **DERIVED-CONDITIONAL** ($\alpha_{UV} = 1$); T3 DERIVED-COND; **T4a PROVEN (theta rigidity)**, **T4b/T4c DERIVED-CONDITIONAL (register-trace)**; **T5 register-tree DBI, full 1PI C_{UV} OPEN**; Mod PROVEN; T7 metric scale OPEN under second-scale dynamics.

Step 4 (convergence). Re-traversal change counts [5, 1, 0] strictly decreasing $\Rightarrow$ converged. v1.2 note: the parent-action expansion **discharged** two previously-conditional nodes (partition non-circularity $\rightarrow$ ZS-M6; anomaly ansatz $\rightarrow$ forced E_λ) and **grounded** one (bulk-vanishing $\rightarrow$ ZS-M6 §10.3) *without*

opening new OPEN nodes — a genuine convergence check (the decomposition tightened, it did not diverge).

Step 5 (value). Versus v1.1: $c_e = 2\pi$ DERIVED $\rightarrow$ PROVEN; anomaly cycle forced; $\tilde{G}_s = 1$ DERIVED-COND $\rightarrow$ DERIVED-BY-INHERITANCE; $\Delta_{bulk} = 0$ register-grounded; the scale no-go sharpened to the loxodromic length ℓ . Converged + corpus-non-collision (M1/M6/F33/A27/A28) + anti-numerology passed. **This is convergence with three frontier nodes precisely OPEN** — physical c_e (conditional on $\alpha_{UV} = 1$), register-trace normalization / C_{UV} (full 1PI OPEN), and the metric scale M_K/M_P (OPEN under second-scale dynamics). Terminus: an **honest conditional template**, not a closure — matching the corpus pattern that internal iteration converges to honesty, not to closure.

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 16. ZS-A27 v2.3, *The Z-Spin Vacuum-Energy Scale — A Dimensional No-Go, the Four-Root Map, and the Closure Frontier* (Z-Spin corpus).
 17. ZS-A28 v2.0, *Vacuum Energy as a Projector-Valued Top Form — Conditional Closure of B3-D* (Z-Spin corpus).
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§19. Version History

- **v1.0 (July 2026):** NG1, T1 ($+K_\Sigma = A_\Sigma^{-1}$), T2, anomaly gates, residual collapse, scale no-go, conditional closure. 17/17 checks.
- **v1.1 (July 2026):** Restored the two load-bearing computations under-developed in v1.0: full-flux partition matching (F36.T4) and the 1PI threshold audit (F36.T5); added the Σ_2 cohomological no-go and the necessary-and-sufficient $c_e = 2\pi \Leftrightarrow \alpha_{UV} = 1$. 30/30 checks.
- **v1.2 (July 2026):** *Parent-action deep exploration executed.* **(1)** Fixes the internal cycle to the **corpus-forced Koenigs torus** $\Sigma_2 = E_\lambda^*$ (F34/F33.2B), correcting v1.1's free S^2 ansatz. **(2)** Computes the wrapped-brane **charge pairing** on T^2 (cellular complex; Betti (1, 2, 1); torsion-free; flux-pairing Smith normal form [1]; unimodular intersection form), upgrading $c_e = 2\pi$ from DERIVED to **PROVEN**. **(3)** Computes the **anomaly gates** on $E_\lambda^*(T^2)$ parallelizable $\Rightarrow w = 1, W_3 = 0$, torsion = 0, $p_1 = 0$), removing the ansatz-dependence (now conditional only on a trivial normal bundle). **(4)** Discharges the partition-matching **non-circularity** via **ZS-M6** (Register-Total Normalization $\kappa^2 = A/Q + \text{Dimensional Coupling Norm } g_{reg}^2 = \dim(Y)\kappa^2$), upgrading $\tilde{G}_s = 1$ to **DERIVED-BY-INHERITANCE** (under ZS-M6 R-1/R-2/R-3). **(5)** Grounds the threshold **bulk-vanishing** in the ZS-M6 §10.3 odd-mode decoupling ($\Delta\lambda(\text{slot } 1) = 0$). **(6)** Sharpens the scale no-go: the Koenigs **shape** $\tau = (\omega + i\kappa_\lambda)/2\pi$ is M1-fixed, but the physical area $A_\Sigma = \ell^2 \text{Im } \tau$, τ needs the loxodromic length $\ell = M_{UV}^{-1}$ — the missing datum, now explicit. **(7)** Executes the six pre-registered companion modules (charge pairing, anomaly gates, register projection, Koenigs modulus, radion, firewall), previously deferred. Verification $30 \rightarrow \mathbf{56/56}$. Zero fitted parameters; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED. **(v1.2 was later found to over-claim; see v1.8.)**
- **v1.3 (July 2026):** *Propagates the ZS-M44 result.* The companion Math-Spine paper ZS-M44 (*The Register-Trace Normalization Theorem*) closes ZS-M6's **R-1** caveat — proving the “/Q” in

$\kappa^2 = A/Q$ is the a_1 Seeley–DeWitt mode count (PROVEN = Q , coupling-immune), not a heuristic — and derives **R-3** from ZS-M6 §7A. Consequently the $\tilde{G}_s = 1$ status (§6) tightens from “DERIVED-BY-INHERITANCE under R-1/R-2/R-3” to “**under the residual R-2 only**” (the register-scalar lattice embedding). Ledger T4, §14, and the abstract updated accordingly. **No numerical value changes**, and the metric-scale no-go (M_{UV} PROVEN-irreducible) is untouched — R-1/R-2/R-3 concern only the dimensionless coupling. Verification unchanged at 56/56 (ZS-M44 carries its own 13/13).

- **v1.4 (July 2026): Propagates ZS-M44 v1.1 (R-2 closure).** The companion paper now discharges **all three** ZS-M6 caveats: R-1 (a_1 mode-count normalization), R-3 (§7A), and **R-2** — split into R-2(a) within-irrep constancy (DERIVED, Schur + polyhedral symmetry) and R-2(b) cross-sector democratic normalization (CONFIRMED to 0.45% by an exact arrowhead inverse-eigenvalue solve, excluding sector-weighting $\delta_x/\delta_y = 0.865$, and correcting the ZS-M6 §10.4 “~15% asymmetry” as a 2nd-order perturbation artifact). Consequently $\tilde{G}_s = 1$ (§6) tightens from “under R-2 only” to “**all named caveats discharged**” (DERIVED-BY-INHERITANCE); the sole residual is a verification-level 50-digit spectral recompute. Ledger T4 (82→88%), abstract, §14 updated. **No numerical value changes**; the metric-scale no-go (M_{UV} irreducible) is untouched. Verification 56/56 (ZS-M44 v1.1 carries 20/20).
- **v1.3–v1.7 (July 2026, superseded):** Propagation drafts (ZS-M44 v1.1→v1.2, ZS-M45, ZS-A31) that progressively claimed $\tilde{G}_s = 1$ had “no undischarged rigor item.” **These over-claims are withdrawn in v1.8.**
- **v1.6 (July 2026): Propagates ZS-M45 + ZS-A31. ZS-M45** closes the last M-sector item (the cell-by-cell lattice- $\hat{R}$): the ZS-M3 uniform holonomy deficit $\delta\phi_{cell} = A(I_{cell} = \kappa/r = 4/4 = 1$, sector-independent) realizes the register-scalar curvature operator without Schur, so $\tilde{G}_s = 1$ has no residual rigor item. **ZS-A31** consolidates the metric scale: the F33–F36 line reduces to one free dimensional parameter M_{UV} ($\rho_{\Lambda Z}/M_{UV}^4 = \frac{1}{2}(1260/4807)\omega^2 = 0.669$ DERIVED; $M_{UV} \approx 2.48$ meV, $\ell \approx 80$, μm by regression; irreducible by ZS-A17/A27/A28; the H_0/M_p debt OPEN per A22 B3). **No numerical value changes**; the metric-scale no-go is untouched. Verification 56/56 (ZS-M45 6/6, ZS-A31 6/6).
- **v1.7 (July 2026): Notes the ZS-A31 v1.1 B3 deep exploration.** The last genuine OPEN of the F33–F36 line — the $H_0/M_p \sim 10^{-61}$ action-level mechanism (A22 barrier B3) — was deep-explored in ZS-A31 v1.1 §8. Verdict: **still OPEN** (no mechanism in the current corpus), but materially sharpened: (i) B3 reduces to the single number $M_{UV}/M_p \approx 2.0 \times 10^{-31}$; (ii) it is a **debt, not a no-go** (the ratio is dimensionless; ZS-A27/A28 do not strictly forbid it); (iii) the coincidence $M_{UV}/M_p \approx A^{|V_Y - F_Y|} = A^{28}$ (1.7%) is registered HYPOTHESIS-weak (30%, no

mechanism, anti-numerology caveats explicit); (iv) closure needs the ZS-A17 emergent-metric advance, beyond current tools. **No F36 numerical value changes**; the metric-scale no-go is untouched. Verification 56/56 (ZS-A31 v1.1 carries 12/12).# **The Integral UV Normalization of the Z-Spin Odd Three-Form: Exact $C_5 \rightarrow A_3$ Reduction, Wrapped-Brane Charge Pairing on the Koenigs Torus, and the Metric-Scale Gate**

- v1.8 (July 2026, review-corrected):** Integrates external review and cleans the version structure. **Honest downgrades:** (1) $\tilde{G}_s = 1 \rightarrow$ **DERIVED-CONDITIONAL on register-trace normalization**, following the ZS-M44 v1.2 a_0/a_1 correction (the “/Q” is the normalized-trace rank $a_0 = \dim = Q$, not $a_1 = Tr, L$; the density $\rho = I_Q/Q$ is asserted, not action-derived) and ZS-M45 v1.1 (uniform cell deficit gives cell-coefficient uniformity, not the register-scalar operator $\hat{R} \propto I_Q$). (2) Dimensionful $c_e = 2\pi \rightarrow$ **DERIVED-CONDITIONAL on $\alpha_{UV} = 1$** (only the primitive charge = 1 and dimensionless 2π are PROVEN). (3) $C_{norm} = 1 \rightarrow$ register tree branch **DBI**, full parent 1PI **OPEN** ($F_4 \wedge \star F_4$ gauge-invariant and seam-even, so not forbidden by gauge invariance + seam parity). (4) Honest terminus $\chi_-^{(s)} = (1260/4807)$, $C_{UV} M_K^4$ with the **metric-scale gate OPEN**; the closing path is the modular-depth $e^{-2\pi t_*}$ frontier (ZS-M46/M47/A31 v1.2). Ledger T4 (90→72%), T2-NS, T5 downgraded; verification 56/56 + review 14/14. **No numerical value changes**; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED. **F36 now terminates honestly at “metric-scale gate OPEN,” pending ZS-M46/M47.**
- v1.9 (July 2026, current — body synchronized):** Second review found v1.8’s metadata honest but the body still carrying v1.2 closure language. v1.9 synchronizes throughout: (a) abstract “computes ... from one parent action” $\rightarrow$ “organizes ... within a common parent-action template”; (b) T4 split into **T4a theta-rigidity (PROVEN) / T4b physical partition matching (DERIVED-CONDITIONAL) / T4c $\tilde{G}_s = 1$ (DERIVED-CONDITIONAL)**; (c) §5 charge pairing $\rightarrow q_{min} = 1$ and dimensionless 2π **PROVEN**, dimensionful $c_e = 2\pi$ **DERIVED-CONDITIONAL on $\alpha_{UV} = 1$** ; (d) §7 T5 body no longer reverses the top verdict — $C_{norm}^{reg,tree} = 1$ **DBI**, C_{UV}^{1PI} **OPEN**, Δ_{brane} **EXPECTED not COMPUTED**; (e) §10 separates the physical general form $\chi_-^{(s)} = (1260/4807)C_{UV} M_K^4$ from the canonical reparameterization $M_{eff} := C_{UV}^{1/4} M_K$; (f) $w = -1 \rightarrow$ **DERIVED-BY-INHERITANCE within the frozen branch** (not IMPORTED-PROVEN), with updated DESI DR2 language ($\sim 3.1\sigma$ BAO+CMB, $\sim 2.8-4.2\sigma$ with SNe); (g) Appendix epistemic tree updated to **convergence with three frontier OPEN nodes** (physical c_e , register-trace/ C_{UV} , metric scale). Honest terminus: $\chi_-^{(s)} = (1260/4807)C_{UV} M_K^4$, **metric-scale gate OPEN**; path via modular depth (ZS-M46/M47/A31). **No numerical value changes**; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED. Cross-refs: ZS-M44 v1.4 (integrates ZS-M45; executes §6.3).

- v1.9 (July 2026):** Synchronized the body to the v1.8 metadata (T4 split into T4a PROVEN / T4b–T4c conditional; c_e split; T5 register-tree DBI vs full-1PI OPEN; $w = -1$ frozen-branch DBI; Appendix tree = convergence with three frontier OPEN nodes). (*Some ledger/anomaly conflicts remained — fixed in v2.0.*)
- v2.0 (July 2026, terminal):** Third review integrated — removes the last status conflicts. Abstract/§6 no longer say “ $\tilde{G}_s = 1$ DERIVED-BY-INHERITANCE”/“non-circularity discharged” (now: algebraic back-definition avoided, register-trace bridge conditional). Claim-ledger **CP** split (Smith form/ q_{min} /dimensionless 2π PROVEN; physical $c_e = 2\pi$ DERIVED-CONDITIONAL); **Coll** row corrected (convention split collapses to $\tilde{g}^2$, but (C_{UV}, M_K) remain); canonical $\chi_-^{(s)} = (1260/4807)M_{eff}^4$ relabeled **IDENTITY / reparameterization**. **Anomaly** condition strengthened: internal $p_1(T^2) = 0$ gives no *intrinsic* shift, but the full flux-quantization shift is **DERIVED-CONDITIONAL** on the ambient shifted classes $(\lambda_{grav}(Y_6)|_{W_5}, [H]|_{W_5}, W_3(NW_5))$. $b_2 = 1$ reworded as “rank-one H_2 generator, unique up to homology” (not “the only 2-cycle”). Cross-refs updated to ZS-M44 v1.5, ZS-A31 v1.4. **No numerical value changes**; $(A, Q, dimZ) = (35/437, 11, 2)$ LOCKED. **Terminal version: the three OPEN items** (C_{UV} , register measure, M_K/M_P^-) are separate research programmes.
- v2.1 (July 2026, terminal — final synchronization):** Removes the last four residual old-version statements flagged in the fourth review; **no new research, no numerical value changes** ($(A, Q, dimZ) = (35/437, 11, 2)$ LOCKED; verification identical, 56/56 + 14/14). **(1)** §5 anomaly body reconciled with the v2.0 abstract: the **A1** clause now states that the internal $p_1(T^2) = 0$ removes only the *intrinsic* internal-tangent contribution, while unshifted flux quantization additionally requires the ambient class $\lambda_{grav}(Y_6)|_{W_5} = 0$; the “Upgrade over v1.1” residual is restated as the full ambient + normal-bundle set $(\lambda_{grav}(Y_6)|_{W_5} = 0, [H]|_{W_5} = 0, W_3(NW_5) = 0)$, and the **claim-ledger T3** condition column updated to match (was “trivial normal bundle”). **(2)** §8: the $M_{eff} := C_{UV}^{1/4} M_K$ definition no longer says “Adopting $C_{UV} = 1$ ” — the definition is C_{UV} -agnostic. **(3)** §10: the regression sentence now fixes only $M_{eff} \approx 2.5$ meV (ZS-A31 v1.4), with $\ell = M_K^{-1} \approx 80, \mu\text{m}$ flagged **DERIVED-CONDITIONAL on** $C_{UV} = 1$; the redundant “ $\chi_-^{(s)} = (1260/4807)M_{UV}^4$ DERIVED” ledger row is deleted, and the closing rows now read $M_{eff} \approx 2.5$ meV (REGRESSION) and $\ell \approx 80, \mu\text{m}$ (DERIVED-CONDITIONAL on $C_{UV} = 1$). **(4)** Abstract A31 hand-off aligned to ZS-A31 v1.4: $\rho_{\Lambda, Z}/M_{eff}^4 = \frac{1}{2}(1260/4807)\omega^2$ (identity), hierarchy frontier $M_K/M_P^- = ? e^{-2\pi Q}$. Per the review directive, **F36 is now closed to further**

revision; the three genuine OPEN items (C_{UV} full-1PI, action-derived register measure, M_K/M_P^-) remain separate research programmes.

- **Version:** v2.1 (supersedes v1.0–v2.0; integrates external review; version structure cleaned). **v2.1 completes the honest synchronization** (the third and fourth reviews found and cleared the residual T4/CP/collapse/anomaly status conflicts). This is the **terminal version** of the F33–F36 charge-unit line; the three genuine OPEN items (C_{UV} full-1PI, action-derived register measure, metric scale M_K/M_P^-) are each a *separate research programme*, not a further F36 revision. Now uniformly: (i) following the ZS-M44 v1.2 a_0/a_1 correction and ZS-M45 v1.1, $\tilde{G}_s = 1$ is **DERIVED-CONDITIONAL on register-trace normalization**, not “no undischarged rigor item”; (ii) the *dimensionful* $c_e = 2\pi$ is **DERIVED-CONDITIONAL on $\alpha_{UV} = 1$** (only the primitive charge = 1 and the *dimensionless* WZ phase 2π are PROVEN); (iii) $C_{norm} = 1$ holds at the **register tree level (DBI)** but the **full parent 1PI matching is OPEN** — $F_4 \wedge \star F_4$ is gauge-invariant and seam-even, so gauge invariance + seam parity do not forbid the counterterm. The honest terminus is $\chi_-^{(s)} = (1260/4807)$, C_{UV}, M_K^4 with the **metric-scale gate OPEN**; the path to close it (modular depth $e^{-2\pi t_*}$) is scoped to ZS-M46/M47/A31.